

THE SCHUBERT BIALGEBRA AND THE RING OF RC GRAPHS

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1. SOME PRELIMINARIES

1.1. Notation.

Definition 1.1.1. For a permutation v' such that $v' \in S_n$, define

$$\varphi_{i,n}(v')(j) = \begin{cases} v'(j) & \text{if } j < i \\ n+1 & \text{if } j = i \\ v'(j-1) & \text{if } i < j \leq n+1 \\ j & \text{if } j > n+1 \end{cases}$$

Now fix $v \in S_n$ and $i \geq 1$ an integer, we define a relation $\searrow^i$ by declaring that $v \searrow^i v'$ if

$$v \xRightarrow{i} \varphi_{i,n}(v')$$

Equivalently, $v \searrow^i v'$ if whenever $v \in S_n$ and n is minimal, we have that v satisfies the relation $\xRightarrow{i}$ with respect to the permutation obtained from v' by inserting $n+1$ at position i . We note that this concept was introduced by Bergeron and Sottile in [4].

If $v \searrow^i v'$, we define a set of integers $Q_i(v', v)$ by

$$Q_i(v', v) = \{v(j) \mid j > i \text{ and } v'(j-1) = v(j)\}$$

Given the fact that any element of S_∞ fixes all but finitely many positive integers, it follows that $Q_i(v', v)$ is a finite set.

We then define $\mathcal{D}_i(v)$ to be all permutations $v' \in S_\infty$ such that $v \searrow^i v'$. The permutations $\mathcal{D}_i(v)$ arise when pulling a variable out of a Schubert polynomial and expressing the coefficients of the powers of this variable as Schubert polynomials in the remaining variables, as is done in [3], from which this definition essentially comes.

For a permutation $v \in S_\infty$, we define $\uparrow v$ to be the permutation defined by

$$\uparrow v(i) = \begin{cases} v(i-1) + 1 & \text{if } i > 1 \\ 1 & \text{if } i = 1 \end{cases}$$

Recursively, we define $\uparrow^k(v)$ to be $\uparrow(\uparrow^{k-1}(v))$.

For a permutation w , we define $\mathbf{m}(w)$ to be the maximum right descent of w . That is

$$\mathbf{m}(w) = \max\{i \mid w(i) > w(i+1)\}$$

Define

$$\Pi(a \mid B) = \prod_{b \in B} (a - b)$$

Proposition 1.1.2 (Special case of Pieri formula [11, Theorem 7.1]). *Suppose $k \geq 1$ and $u, w \in S_\infty$. If $u \xrightarrow{k} w$, then*

$${}_{(u)}\partial_u^w \Pi(x_1 \mid y_{[k]}) = 0$$

If $u \xrightarrow{k} w$, define

$$Q = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$

then

$${}_{(y)}\partial_u^w \Pi(x_1 \mid y_{[k]}) = \Pi(x_1 \mid y_Q)$$

Proof. This is simply a change of variables from the original theorem. $\square$

The next proposition specializes, in the case of ordinary Schubert polynomial $\mathfrak{S}_v(x)$, to a formula that isolates a chosen index i : one can express $\mathfrak{S}_v(x)$ as a sum of terms of the form $x_i^p \mathfrak{S}_{v'}(x^{(i)})$, thereby effectively extracting the variable x_i and leaving Schubert polynomials in the remaining variables [4, Theorem 5.1]. Proposition 1.1.3 is the double Schubert polynomial version of this, which, as far as we know, is new.

Proposition 1.1.3. *Let $v \in S_\infty$ and let $i > 0$ be an integer. Then we have*

$$\mathfrak{S}_v(x; y) = \sum_{v \xrightarrow{i} v'} \Pi(x_i \mid y_{Q_i(v', v)}) \mathfrak{S}_{v'}(x^{(i)}; y)$$

Proof. Suppose $v \in S_n$. We have

$$\mathfrak{S}_v(x; y) = {}_{(y)}\partial^{vw_0(n)}(\mathfrak{S}_{w_0(n)}(x; y))$$

This is equal to

$${}_{(y)}\partial^{vw_0(n)}(\mathfrak{S}_{s_{n+1-i} \cdots s_1 w_0(n)}(x^{(i)}; y) \Pi(x_i \mid y_{[n+1-i]}))$$

and, applying the Leibniz formula, is also equal to

$$\sum_{\substack{v' \in S_\infty \\ \ell(v' w_0(n) s_1 \cdots s_{n+1-i}) = \ell(w_0(n) s_1 \cdots s_{n+1-i}) - \ell(v')}} \mathfrak{S}_{v'}(x^{(i)}; y) {}_{(y)}\partial_{v' w_0(n) s_1 \cdots s_{n+1-i}}^{vw_0(n)} \Pi(x_i \mid y_{[n+1-i]})$$

By the restricted Pieri formula (Proposition 1.1.2), for this to be nonzero necessarily $v' w_0(n) s_1 \cdots s_{n+1-i} \xrightarrow{n+1-i} v w_0(n)$. We note that $v' w_0(n) s_1 \cdots s_{n+1-i} \xrightarrow{n+1-i} v w_0(n)$ if and only if

$$v \xrightarrow{i} v' w_0(n) s_1 \cdots s_{n+1-i} w_0(n) = v' s_n s_{n-1} \cdots s_i$$

We have that $v' s_n \cdots s_i$ is exactly $\varphi_{i,n}(v')$ since $v' \in S_n$, so we require that

$$v \xrightarrow{i} \varphi_{i,n}(v')$$

so the sum is over all $v' \in \mathcal{D}_i(v)$.

Applying Proposition 1.1.2, we obtain that the result is equal to

$$\sum_{v' \in \mathcal{D}_i(v)} \Pi(x_i \mid y_{A(v', v)}) \mathfrak{S}_{v'}(x^{(i)}; y)$$

where $A(v', v)$ is the set of all $vw_0(n)(j)$ such that $1 \leq j \leq n+1-i$ and $v' w_0(n) s_1 \cdots s_{n+1-i}(j) = vw_0(n)(j)$. These values are the same as at the indices that comprise the set of all $1 \leq j \leq n+1-i$ such that

$$v'(n+2 - s_1 \cdots s_{n+1-i}(j)) = v(n+2 - j)$$

Applying the $s_1 \cdots s_{n+1-i}$ to j , since $1 \leq j \leq n+1-i$ we have that

$$s_1 \cdots s_{n+1-i}(j) = j+1$$

Hence we need

$$v'(n+2 - (j+1)) = v'(n+1 - j) = v(n+2 - j)$$

Replacing j with $n+2 - p$, the indices are the set of all p such that $i < p \leq n+1$ and

$$v'(p-1) = v(p)$$

Thus $A(v', v)$ is the set of all $v(p)$ such that $p > i$ and $v'(p-1) = v(p)$, which is exactly $Q_i(v', v)$, and we are done. $\square$

2. THE SCHUBERT BIALGEBRA AND ITS DUAL

2.1. Definition. We define a commutative algebra $\mathcal{A}$ over the integers as follows. For each n , define $\mathcal{A}_n$ to be the polynomial ring over $\mathbb{Z}$ in the variables $x_1, \dots, x_n$. Then define

$$\mathcal{A} = \bigoplus_{n=0}^{\infty} \mathcal{A}_n$$

The multiplication within $\mathcal{A}_n$ is as usually defined for the polynomial ring. However, if $a \in \mathcal{A}_m$ and $b \in \mathcal{A}_n$ with $m \neq n$ and $m, n > 0$, then

$$ab = 0$$

Otherwise, the component for $n = 0$ is identified with the coefficient ring. Note that the “identity element” for positive n is not an identity element of $\mathcal{A}$ (we may sometimes refer to it as a “fat identity”).

Each $\mathcal{A}_n$ has a basis consisting of elements x_a , where a is a weak composition of length n , and the notation indicates that

$$x_a = x_1^{a_1} \cdots x_n^{a_n}$$

The direct sum therefore has a basis that can canonically be identified with union of these.

We define a coproduct $\Delta : \mathcal{A} \rightarrow \mathcal{A} \otimes \mathcal{A}$ on the basis $x_c^{(n)}$ by

$$\Delta(x_c) = \sum_{ab=c} x_a \otimes x_b$$

where the equation $ab = c$ indicates that a concatenated with b is equal to c . We also define a counit $\varepsilon : \mathcal{A} \rightarrow \mathbb{Z}$ by $\varepsilon(x_a) = 0$ unless a is the empty composition.

Lemma 2.1.1. *With Δ and ε , $\mathcal{A}$ is a coassociative, counital coalegebra.*

Proof. We have

$$\Delta(x_d^{(n)}) = \sum x_a^{(p)} \otimes x_c^{(q)}$$

Applying Δ to either tensor factor results in

$$\sum x_a^{(p)} \otimes x_b^{(q)} \otimes x_c^{(r)}$$

The symmetry of this is exactly the coassociativity condition. Seeing that we may choose $p = n$ or $q = n$, the definition of the counit gives us the result that $\mathcal{A}$ is counital as well under Δ and ε . $\square$

Lemma 2.1.2. *$\Delta : \mathcal{A} \rightarrow \mathcal{A} \otimes \mathcal{A}$ is a homomorphism of rings.*

Proof. This is where the condition that $x_a^{(p)} x_b^{(q)} = 0$ unless $p = q$ when both $p, q > 0$ comes in. It ensures that only monomials of the same length have nonzero products and preserves the structure of the coproduct as a homomorphism of rings. $\square$

Lemma 2.1.3. *$\nabla : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$ is a homomorphism of coalgebras.*

Proof. Consider the basis element $x_a \otimes x_b$ of $\mathcal{A} \otimes \mathcal{A}$. The product is

$$x_{a+b}$$

Applying Δ results in

$$\sum_{a'b'=a+b} x_{a'} \otimes x_{b'}$$

Conversely, applying $\Delta \otimes \Delta$ yields

$$\sum_{a_1 a_2 = a, b_1 b_2 = b} x_{a_1} \otimes x_{a_2} \otimes x_{b_1} \otimes x_{b_2}$$

and applying $\nabla \otimes \nabla$ yields

$$\sum x_{a_1+b_1} \otimes x_{a_2+b_2}$$

The definition of ∇ ensures that a_1 and b_1 are the same length for a nonzero term, and similarly for a_2 and b_2 . Setting $a_1 + b_1 = a'$, $a_2 + b_2 = b'$ yields the result. $\square$

Corollary 2.1.4. *$\mathcal{A}$ is a bialgebra over $\mathbb{Z}$.*

$\mathcal{A}$ is afforded a grading into finite dimensional components by observing that each $\mathcal{A}_n$ itself is a graded ring with each homogeneous component being a finitely generated free module. Considering the pair (n, d) , where n is the number of variables and d is the degree, as a $\mathbb{Z}^2$ grading, we may take the graded dual module $\mathcal{D}$, which, by virtue of the grading, is isomorphic as a free module to $\mathcal{A}$. We identify the dual basis element x_a^* with the sequence of nonnegative integers a . That is to say,

$$\langle a, x_b \rangle = \delta_{ab}$$

Inspection of the coproduct reveals that the product of a and b in $\mathcal{D}$ is simply the concatenation ab . Hence $\mathcal{D}$ is isomorphic to the free associative algebra on a countable set indexed by nonnegative integers.

$\mathcal{D}$ also has a coproduct compatible with its product, namely

$$c \mapsto \sum_{a+b=c} a \otimes b$$

Thus $\mathcal{A}$ and $\mathcal{D}$ are dual bialgebras.

Calling $\mathcal{A}$ the “Schubert bialgebra” may seem unnecessarily grandiose, however the reason will become clear below.

2.2. The Schubert basis. In $\mathcal{A}$, each $\mathcal{A}_n$ has a basis of Schubert polynomials $\mathfrak{S}_u^{(n)}$, where the largest right descent of u is at most n , ensuring that $\mathfrak{S}_u(x)$ has at most n variables. Schubert polynomials limited to a specific number of variables have well-defined structure constants $c_{u,v}^w$, independent of n , given by

$$\mathfrak{S}_u^{(n)} \mathfrak{S}_v^{(n)} = \sum_w c_{u,v}^w \mathfrak{S}_w^{(n)}$$

These are known to be nonnegative integers by virtue of the fact that they count points of transverse intersections of generic translates of Schubert varieties, however except in special cases no positive combinatorial formula is known.

2.3. The elementary basis. An n -elementary weak composition is a finite sequence of nonnegative integers $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_N)$ such that $\alpha_i \leq i$ for all $i < n$ and $\alpha_i \leq n$ for all $i \geq n$, satisfying the additional restriction that $\alpha_i \geq \alpha_{i+1}$ if $i \geq n$.

An n -elementary monomial is an element of $\mathcal{A}_n$ of the following form:

$$e_{\alpha_1}^1 \cdots e_{\alpha_n}^n e_{\alpha_{n+1}}^n \cdots e_{\alpha_m}^n$$

where α is an n -elementary weak composition. Define $\mathcal{E}_n$ to be the set of n -elementary monomials.

Proposition 2.3.1. $\mathcal{E}_n$ forms a $\mathbb{Z}$ -basis for $\mathcal{A}_n$, and $\bigcup_n \mathcal{E}_n$ forms a $\mathbb{Z}$ -basis for $\mathcal{A}$.

Proof. It is a theorem of Macdonald that the polynomial ring $\mathbb{Z}[x_1, \dots, x_n]$ is a free module over Λ_n with basis the Schubert polynomials $\mathfrak{S}_u(x)$ such that $u \in S_n$. These Schubert polynomials are uniquely expressed in terms of strict elementary symmetric monomials with fewer than n variables. Since Λ_n is a polynomial ring in the $e_{i,n}$ by the standard characterization of Λ_n as a polynomial ring in $e_1, \dots, e_n$, Λ_n has a basis of monomials e_λ^n as λ ranges over all partitions with parts bounded by n . The multiplicative combination of these two bases is therefore a $\mathbb{Z}$ -basis of $\mathcal{A}_n$ by freeness of the module. This is exactly the description of the monomials e_α^n for α an n -elementary weak composition. $\square$

We can ask for transition formulas between the Schubert and the elementary basis. From elementary to Schubert, we may use Sottile’s Pieri formula iteratively [12], which provides a positive formula for multiplying a Schubert polynomial by an elementary symmetric polynomial. That is,

$$e_\alpha^n = \sum_{1 \xrightarrow{1,2,\dots,n} \alpha u} \mathfrak{S}_u^n$$

For the reverse transition, we need a lemma. Recall the skew divided difference operators ∂_u^w for $u, w \in S_\infty$ defined originally in [8], though we use the notation of [10], by the Leibniz formula

$$\partial^w(pq) = \sum_u \partial^u(p) \partial_u^w(q)$$

where p and q are arbitrary polynomials and u is a permutation such that $u \leq w$ in Bruhat order. The formula below provides a formula for application of a skew divided difference operator of degree $-n$ to a

polynomial of degree n , which yields an integer for integer-coefficient polynomials. Specifically, it is a formula for application of such an operator with $\ell(u, w) = n$ specifically to x_p^n for some $p > 0$.

Definition 2.3.2. For $u, w \in S_\infty$ such that $u \leq w$, we define $\mathcal{C}^{(p)}(u, w)$ to be the set of saturated chains C in the Bruhat interval $[u, w]$ of the form

$$u = u_0 \prec u_1 \prec \cdots \prec u_n = w$$

such that $u_{i-1}(p) \neq u_i(p)$ for all i . We also define $\sigma(C)$ to be the number of indices i for which the index $j \neq p$ such that $u_{i-1}(j) \neq u_i(j)$ satisfies $j < p$.

Lemma 2.3.3. Let $n \geq 0$ be an integer, let $p > 0$ be an integer, and let $u, w \in S_\infty$ be permutations such that $\ell(u, w) = n$. Then we have that

$$\partial_u^w(x_p^n) = \sum_{C \in \mathcal{C}^{(p)}(u, w)} (-1)^{\sigma(C)}.$$

Proof. Consider the case where $\ell(u, w) = 1$. Then there is a transposition t_{ij} such that $w = ut_{ij}$. Assume that $i < j$. If $p \notin \{i, j\}$, then $\partial_u^w(x_p) = 0$. If $p = i$, then $\partial_u^w(x_p) = \partial^{ij}(x_p) = 1$. If instead $j = p$, then $\partial_u^w(x_p) = \partial^{ij}(x_p) = -1$. In all cases, the result holds.

Now, consider the case where $\ell(u, w) = n > 1$. We have that

$$x_p^n = x_p \cdot x_p^{n-1}$$

Applying the Leibniz formula, we have

$$\partial_u^w(x_p^n) = \sum_{u \prec u'} \partial_u^{u'}(x_p) \partial_{u'}^w(x_p^{n-1})$$

By the above, $\partial_u^{u'}(x_p)$ is nonzero if and only if $u \prec u'$ is a cover in the Bruhat order such that $u(p) \neq u'(p)$. In this case, $\partial_u^{u'}(x_p)$ is either 1 or -1 depending on the number of indices $j < p$ such that $u(j) \neq u'(j)$, which is either 0 or 1, respectively. By induction,

$$\partial_{u'}^w(x_p^{n-1}) = \sum_{C' \in \mathcal{C}^{(p)}(u', w)} (-1)^{\sigma(C')}.$$

Hence,

$$\partial_u^{u'}(x_p) \partial_{u'}^w(x_p^{n-1}) = \begin{cases} - \sum_{C' \in \mathcal{C}^{(p)}(u', w)} (-1)^{\sigma(C')} & \text{if } u(j) \neq u'(j) \text{ for some } j < p \\ \sum_{C' \in \mathcal{C}^{(p)}(u', w)} (-1)^{\sigma(C')} & \text{otherwise,} \end{cases}$$

which, by definition, is

$$\sum_{C = C' \cup \{u\}, C' \in \mathcal{C}^{(p)}(u', w)} (-1)^{\sigma(C)}$$

Summing over all u' such that $u \prec u' \leq w$, we obtain the result. $\square$

Using a minor sleight of hand, this affords us an explicit formula for transition of a Schubert polynomial into the elementary basis.

Definition 2.3.4. Let δ^u be the divided difference operator associated to u on the y variables for a given $u \in S_\infty$, and let

$$E_k(x; y_p) = \prod_{i=1}^k (x_i - y_p)$$

which is a factorial elementary symmetric polynomial, a special case of a double Schubert polynomial.

Lemma 2.3.5. Let $u, w \in S_\infty$ satisfy $u \leq w$ and let $p > 0, k > 0$ be integers. Then we have

$$\partial_u^w(E_k(x; -y_p)) = \sum_{C \in \mathcal{C}^{(p)}(u, w)} (-1)^{\sigma(C)} e_{k-\ell(u, w)}^k + \mathcal{O}(y)$$

Proof. This follows from the observation that

$$E_k(x; -y_p) = \sum_{i=0}^k e_{k-i}^k y_p^i$$

and Lemma 2.3.3. □

Definition 2.3.6. Suppose we want to express $\mathfrak{S}_w^n$ in the elementary basis. Let w_λ be a dominant permutation such that

$$\mathfrak{c}^*(w_\lambda) = \lambda = (n, n, \dots, n, n-1, \dots, 1)$$

and consider the double Schubert polynomial

$$\mathfrak{S}_{w_\lambda}(x; -y) = \prod_i E_{\lambda_i}(x; -y_i)$$

(this equation for $\mathfrak{S}_{w_\lambda}$ follows from [8, (6.14)]).

We have that

$$\mathfrak{S}_w(x; y) = \delta^{w\lambda^{-1}} \mathfrak{S}_{w_\lambda}(x; y)$$

We may apply the divided difference $\delta^{w\lambda^{-1}}$ to the product $\prod_i E_{\lambda_i}(x; -y_i)$ using the Leibniz formula and set $y_i = 0$ for all i , which gives us, by Lemma 2.3.5,

$$\sum_{\substack{P=(C_1, \dots, C_{\ell(\lambda)}) \\ C_i \in \mathcal{C}^{(i)}(u_{i-1}, u_i)}} (-1)^{\sigma(C_1) + \dots + \sigma(C_{\ell(\lambda)})} e_{\lambda_1 - |C_1|}^{\lambda_1} e_{\lambda_2 - |C_2|}^{\lambda_2} \cdots e_{\lambda_{\ell(\lambda)} - |C_{\ell(\lambda)}|}^{\lambda_{\ell(\lambda)}}$$

where $1 = u_0 \leq u_1 \leq \dots \leq u_{\ell(\lambda)} = w\lambda^{-1}$. Define, for convenience,

$$\sigma(P) = \sum_{i=1}^{\ell(\lambda)} \sigma(C_i)$$

for such a P . Then we define

$$E_w^{\alpha; n} = \sum_{\substack{P=(C_1, \dots, C_{\ell(\lambda)}) \\ C_i \in \mathcal{C}^{(i)}(u_{i-1}, u_i)}} \sum_{(\lambda_{\ell(\lambda)+1-i} - |C_{\ell(\lambda)+1-i}|)_{i=1}^{\ell(\lambda)} = \alpha} (-1)^{|\lambda| - \ell(w) + \sigma(P)}$$

Theorem 2.1 (Transition to the elementary basis). *For each w , we have*

$$\mathfrak{S}_w^n = \sum_{\alpha} E_w^{\alpha; n} e_{\alpha}^n$$

These numbers are stable for fixed w as soon as the number of variables is at least as large as the value of n such that $w \in S_n$, and are well-studied but poorly understood.

Example 2.3.7. Let $n = 5$ and let w be the permutation such that $\mathfrak{c}(w) = (0, 2, 0, 2, 3)$. Then

$$\begin{aligned} \mathfrak{S}_w^n = & e_{(0,0,0,0,5,1,1)} - e_{(0,0,0,0,4,2,1)} + e_{(0,0,0,0,4,3)} - e_{(0,1,0,0,4,1,1)} + e_{(0,0,1,0,3,2,1)} \\ & + e_{(0,1,0,0,5,1)} + e_{(0,1,0,0,3,3)} \end{aligned}$$

Remark. This transition formula can be used to obtain an extremely efficient formula for multiplication of Schubert polynomials. If we want to compute $\mathfrak{S}_u^n \mathfrak{S}_v^n$, we can first express $\mathfrak{S}_v^n$ in the elementary basis using Theorem 2.1. Then we can multiply by $\mathfrak{S}_u^n$ by iterating the Pieri formula. This is the method used in the python package `schubmult` written by the author. In fact, there is also a version for double Schubert polynomials, quantum Schubert polynomials, and quantum double Schubert polynomials, which will not go into here but are implemented in the same package.

3. THE DUAL ALGEBRA

We examine the graded dual algebra $\mathcal{D}$ more closely now. This is a graded ring generated by countably many elements that we denote by $[i]$, where i is a nonnegative integer. The product, as mentioned previously, is concatenation of sequences. Thus

$$[a_1 \cdots a_p][b_1 \cdots b_q] = [a_1 \cdots a_p b_1 \cdots b_q]$$

It is not hard to see that $\mathcal{D}$ is a free algebra on these generators. Thus the set of sequences $[a_1 \cdots a_n]$ forms a $\mathbb{Z}$ -basis for $\mathcal{D}$. We may realize this as the dual of $\mathcal{A}$ by declaring that

$$\langle [a_1 \cdots a_n], x_1^{b_1} \cdots x_n^{b_n} \rangle = \prod_i \delta_{a_i, b_i}$$

We have a $\mathbb{Z} \times \mathbb{Z}$ grading such that

$$\deg([a_1 \cdots a_n]) = (n, -a_1 - a_2 \cdots - a_n)$$

The set of elements such that $\deg(a) = (n, -)$ is $\mathcal{D}_n$, dual as a graded module to $\mathcal{A}_n$.

3.1. The dual Schubert basis. There is a basis Ξ_u^n dual to the Schubert basis for $\mathcal{D}_n$. Specifically, with the unique pairing $\langle -, - \rangle : \mathcal{D} \times \mathcal{A} \rightarrow \mathbb{Z}$ such that

$$\langle \alpha, x_\beta \rangle = \delta_{\alpha\beta}$$

we define Ξ_u^n to be the unique basis of $\mathcal{D}_n$ such that

$$\langle \Xi_u^n, \mathfrak{S}_v^n \rangle = \delta_{uv}$$

We characterize it with an explicit formula.

Theorem 3.1. *For each permutation u and integer n with $\ell(us_i) > \ell(u)$ for all $i > n$ we have the equation*

$$\Xi_u^n = \sum_{\ell(\alpha)=n} E_{u\mu^{-1}}^{\mathfrak{c}(\mu)-\alpha, \mathfrak{c}(\mu)} \alpha$$

where μ is any strict dominant permutation such that $0 \neq \mathfrak{c}_n(\mu) \geq \ell(u)$ and $\mathfrak{c}_{n+1}(\mu) = 0$.

Proof. By definition, the coefficient of α in Ξ_u^n is the coefficient of $\mathfrak{S}_u^n$ in x_α . This can be derived from the Cauchy formula for double Schubert polynomials. Note that for any permutation μ as laid out in the statement of the theorem, $\ell(u\mu^{-1}) = \ell(\mu) - \ell(u)$. Thus,

$$\partial_y^{u\mu^{-1}} \mathfrak{S}_\mu(x; -y) = \mathfrak{S}_u(x; -y)$$

We have that

$$\mathfrak{S}_\mu(x; -y) = \sum_u \mathfrak{S}_u(x) \mathfrak{S}_{u\mu^{-1}}(y)$$

Expressing the second factor in the $e_{\alpha, \lambda}(y)$ basis, we have

$$\mathfrak{S}_\mu(x; -y) = \sum_{u, \alpha} \mathfrak{S}_u(x) E_{u\mu^{-1}}^{\mathfrak{c}(\mu)-\alpha, \mathfrak{c}(\mu)} e_{\mathfrak{c}(\mu)-\alpha, \mathfrak{c}(\mu)}(y)$$

An alternative expression for $\mathfrak{S}_\mu(x; -y)$ is

$$\mathfrak{S}_\mu(x; -y) = \sum_\alpha x_\alpha e_{\mathfrak{c}(\mu)-\alpha, \mathfrak{c}(\mu)}(y)$$

from which we see that the coefficient is correct. □

This is not stable for fixed u as n increases, and this is expected.

Example 3.1.1. Suppose $w = [1, 4, 5, 2, 3]$. Then

$$\Xi_w^3 = [022] - [013]$$

If we chose $[1, 3, 5, 7, 2, 4, 6]$ instead, we have

$$\Xi_{1357246} = [0015] - [0033] - [0114] + [0123]$$

For $w = [4, 2, 7, 1, 3, 5, 6]$,

$$\Xi_{4271356}^3 = -[026] + [035] + [125] - [134] + [206] - [215] - [305] + [314]$$

Thus the product structure of $\mathcal{D}$ encodes splitting of the Schubert polynomial into two sets of variables. In particular,

Lemma 3.1.2. *Let $a \geq 0$ be an integer and $w \in S_\infty$. Then we have*

$$[a] \cdot \Xi_w^n = \sum_{\substack{w' \in \mathcal{D}_1(w') \\ \ell(w, w') = a}} \Xi_{w'}^{n+1}$$

4. THE RING OF BOUNDED RC GRAPHS

4.1. The module of bounded RC graphs.

Definition 4.1.1. Let $R \subseteq \mathbb{P} \times \mathbb{P}$ be a finite set. To each such set we associate an element w_R of S_∞ as follows. Given a pair (i, j) where $i, j > 0$ are integers, define $s(i, j) = s_{i+j-1}$. Totally order the grid such that $(i, j) < (a, b)$ if and only if $i < a$ or $i = a$ and $j > b$ (in other words, lexicographical order, except that the order on the second coordinate is reversed). By this ordering, index R as $r_1, r_2, \dots, r_m$ in increasing order. Then

$$w_R = s(r_1) \cdots s(r_m)$$

If $\ell(w_R) = m$, then we say that R is an *RC-graph*.

A *bounded RC-graph* is an RC-graph R together with a specified number of rows $n \geq \max\{i : (i, j) \in R \text{ for some } j\}$ such that w_R has no right descent larger than n . The definition is as an ordered pair (R, n) , but it will be far more convenient to abuse notation and consider the number of rows a property of R itself. To denote the number of rows, we define the *height* of R to be $\text{ht}(R) = n$. A bounded RC graph has an associated weight vector $\text{wt}(R)$ where $\text{wt}_i(R)$ is the number of elements in row i .

To a bounded RC graph R with $\text{ht}(R) \geq 1$, we define the *trim* operation $\text{trim}(R)$ to be the bounded RC graph with height $\text{ht}(R) - 1$ and underlying set $\{(i-1, j-1) : (i, j) \in R, i \geq 2\}$. We also define

$$\uparrow^m R = \{(i+m, j) : (i, j) \in R\}$$

Let $\mathcal{BRC}$ be the free abelian group spanned by all bounded RC graphs. If $\mathcal{BRC}_n$ is the subgroup spanned by all bounded RC graphs R with $\text{ht}(R) = n$, then we have a grading

$$\mathcal{BRC} = \bigoplus_{n=0}^{\infty} \mathcal{BRC}_n$$

There is an evident evaluation map $\phi : \mathcal{BRC} \rightarrow \mathcal{A}$ defined on basis elements as

$$\phi(R) = x_{\text{wt}(R)}$$

which is a surjective homomorphism of graded modules. There is also a map $\alpha : \mathcal{BRC} \rightarrow \mathcal{D}$ defined by

$$\alpha(R) = \Xi_{w_R}^{\text{ht}(R)}$$

which is also a surjective homomorphism of graded modules. In addition, there is $\omega : \mathcal{BRC} \rightarrow \mathcal{D}$ defined by

$$\omega(R) = [\text{wt}(R)]$$

which is similarly surjective.

We can define elements $\mathcal{S}_w(n)$ as

$$\mathcal{S}_w(n) = \sum_{\substack{w_R = w \\ \text{ht}(R) = n}} R$$

For a generating element $[a]$ of $\mathcal{D}$ and a bounded RC graph R , we define

$$[a] \cdot R = \sum_{\substack{\text{trim}(R') = R \\ \text{wt}_1(R') = a}} R'$$

which is an element of $\mathcal{BRC}$. By virtue of the fact that $\mathcal{D}$ is a free algebra generated by these elements, it is nearly a trivial observation that this is a left module action on $\mathcal{BRC}$. The consequences of this, however, are not at all trivial.

Lemma 4.1.2 ([2, Corollary 3.11]). *We have that $\mathcal{D}_1(w)$ is equal to the set of permutations w' such that there exists a permutation $v = s_{a_1} \cdots s_{a_m}$ for some integers $a_1 > a_2 > \cdots > a_m \geq 1$ such that $w = v \uparrow w'$.*

Lemma 4.1.3. *Let v be a permutation. If $v \searrow^1 v'$, let $a_1, \dots, a_k$ be the elements of $Q_1(v', v)$ in decreasing order. Then*

$$v = s_{a_1} \cdots s_{a_k} \uparrow v'$$

Proof. Suppose $v \in S_n$. We have by definition that there is a sequence of integers $b_1, \dots, b_p$, all distinct and greater than 1, such that

$$vt_{1,b_1} \cdots t_{1,b_p}(1) = n + 1$$

and

$$vt_{1,b_1} \cdots t_{1,b_p}(i + 1) = v'(i)$$

for all $i < n - 1$. This means that

$$vt_{1,b_1} \cdots t_{1,b_p} = s_n s_{n-1} \cdots s_1 \uparrow v'$$

This is because if $v'' = \uparrow v'$, then

$$v''(1) = 1$$

and

$$v''(i) = v(i - 1) + 1$$

The cycle $s_n \cdots s_1$ sends $1 \mapsto n + 1$ and $i \mapsto i - 1$ if $1 < i \leq n + 1$, hence

$$vt_{1,b_1} \cdots t_{1,b_p} = s_n s_{n-1} \cdots s_1 \uparrow v'$$

In particular,

$$v = s_n s_{n-1} \cdots s_1 \uparrow v' t_{1,b_p} \cdots t_{1,b_1}$$

This results in the factorization

$$v = s_n \cdots s_1 t_{1,v'(b_p-1)+1} \cdots t_{1,v'(b_1-1)+1} \uparrow v'$$

The value $v'(b_j - 1)$ necessarily decreases as j decreases, since applying the corresponding reflection in the reverse order strictly increases the length with each application. Consequently, multiplying $s_n \cdots s_1$ on the right by these reflections removes the simple reflections $s_{v'(b_p-1)}, \dots, s_{v'(b_1-1)}$. The indices removed are precisely the complement of the elements of $Q_1(v', v)$, hence the elements of $Q_1(v', v)$ in decreasing order are what remain, as desired. $\square$

Definition 4.1.4. For a set of positive integers $A = \{a_1, \dots, a_m\}$ with $a_1 > a_2 > \cdots > a_m \geq 1$ and a positive integer i , define

$$\text{row}_i(A) = \{(i, a_j) \mid a_j \in A\}$$

Theorem 4.1. *Let $a \geq 0$ be an integer and let $R \in \mathcal{BRC}$. Then*

$$[a] \cdot R = \sum_{\substack{w_R \in \mathcal{D}_1(w') \\ \ell(w_R, w') = a}} (\text{row}_1(Q_1(w_R, w')) \cup \uparrow R)$$

where the right side denotes bounded RC graphs with $\text{ht}(R) + 1$ rows.

Proof. If $\text{trim}(R') = R$, then by definition $w_{R'} = s_{a_1} \cdots s_{a_m} \uparrow w_R$. It follows by Lemma 4.1.2 then $w_R \in \mathcal{D}_1(w_{R'})$. By Lemma 4.1.3, the integers $a_1, \dots, a_m$ are precisely the elements of $Q_1(w_R, w_{R'})$ in decreasing order. This establishes the result. $\square$

Lemma 4.1.5. *Let $a \geq 0$ be an integer and let $w \in S_\infty$. For any valid n , we have*

$$[a] \cdot \mathcal{S}_w(n) = \sum_{\substack{w \in \mathcal{D}_1(w') \\ \ell(w, w') = a}} \mathcal{S}_{w'}(n + 1)$$

Let t be an indeterminate commuting with all elements of $\mathcal{D}$. Write

$$\mathfrak{S}(t) = \sum_{a=0}^{\infty} [a] t^a$$

Define

$$\mathfrak{S}(x_1, x_2, \dots, x_n) = \mathfrak{S}(x_1) \mathfrak{S}(x_2) \cdots \mathfrak{S}(x_n)$$

Theorem 4.2. Let $\emptyset \in \mathcal{BRC}$ denote the empty bounded RC graph with $\text{ht}(\emptyset) = 0$. Then

$$\mathfrak{S}(x_1, \dots, x_n) \cdot \emptyset = \sum_{w \in S_\infty} \mathfrak{S}_w(x_1, \dots, x_n) \mathcal{S}_w(n)$$

Proof. The result for $n = 1$ is Lemma 4.1.5 together with Theorem 4.1. The general result follows by induction on n . Consider the inductive hypothesis

$$\mathfrak{S}(x_2, \dots, x_n) \cdot \emptyset = \sum_{w \in S_\infty} \mathfrak{S}_w(x_2, \dots, x_n) \mathcal{S}_w(n-1)$$

Then applying $\mathfrak{S}(x_1)$ to both sides, we have

$$\mathfrak{S}(x_1) \mathfrak{S}(x_2, \dots, x_n) \cdot \emptyset = \sum_{w \in S_\infty} \mathfrak{S}_w(x_2, \dots, x_n) \mathfrak{S}(x_1) \mathcal{S}_w(n-1)$$

which is equal to

$$\sum_{a=0}^{\infty} \sum_{w \in S_\infty} x_1^a \mathfrak{S}_w(x_2, \dots, x_n) \mathfrak{S}(x_1)[a] \mathcal{S}_w(n-1)$$

Applying Lemma 3.1.2 to each term, we have

$$\sum_{a=0}^{\infty} \sum_{w \in S_\infty} x_1^a \mathfrak{S}_w(x_2, \dots, x_n) \sum_{\substack{w' \in \mathcal{D}_1(w') \\ \ell(w, w')=a}} \mathcal{S}_{w'}(n)$$

Bringing the polynomial inside the inner sum, this is

$$\sum_{w' \in S_\infty} \left(\sum_{w \in \mathcal{D}_1(w')} x_1^{\ell(w, w')} \mathfrak{S}_w(x_2, \dots, x_n) \right) \mathcal{S}_{w'}(n)$$

which simplifies to

$$\sum_{w' \in S_\infty} \mathfrak{S}_{w'}(x_1, \dots, x_n) \mathcal{S}_{w'}(n)$$

as desired. □

Corollary 4.1.6. For each $w \in S_\infty$ and valid n , we have

$$\phi(\mathcal{S}_w(n)) = \mathfrak{S}_w^n(x_1, \dots, x_n)$$

4.2. The pipe dream visualization and roots. Given a pair of positive integers i, j and an RC graph R , define

$$R[i, j] = \{(a, b) \in R \mid (a, b) > (i, j)\}$$

Given an RC graph R and a pair of positive integers i, j , we define an ordered pair

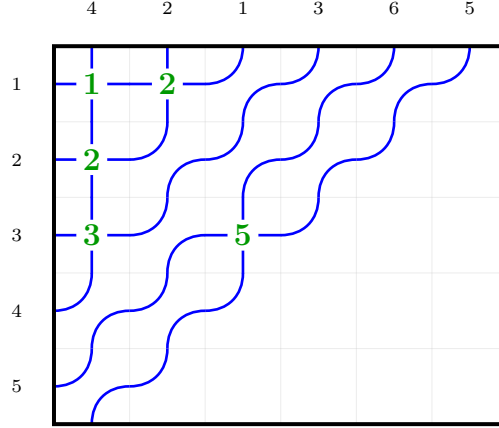
$$\mathbf{rt}_R(i, j) = (w_{R[i, j]}^{-1}(i + j - 1), w_{R[i, j]}^{-1}(i + j))$$

There is a common visualization of RC graphs as pipe dreams. In this visualization, we draw an infinite grid in the first quadrant, and in each position (i, j) we draw either a crossing (if $(i, j) \in R$) or an elbow (if $(i, j) \notin R$). Then we draw pipes entering from the left edge of the grid, with the pipe entering at row i labeled i . The pipes travel through the grid, turning at elbows and crossing at crossings, and exit at the top of the grid, which is labeled with the same number.

A modification to this common visualization that we use has the following additional features:

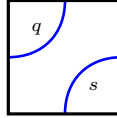
- We write the index of the simple reflections corresponding to the crossings in the grid area itself. Thus, at position (i, j) we write the number $i + j - 1$ if there is a crossing at that position.
- For a bounded RC graph R with $\text{ht}(R) = n$, we only draw the first n pipes entering from the left side of the grid and clip features outside of the first n rows. The pipes exiting at the top are still labeled since the width is not limited.
- *Note:* The labels at the top are not of the permutation w_R , but rather of the permutation w_R^{-1} .

FIGURE 1. The pipe dream visualization of the bounded RC graph R with $\text{ht}(R) = 5$ where $R = \{(1, 1), (1, 2), (2, 1), (3, 1), (3, 3)\}$



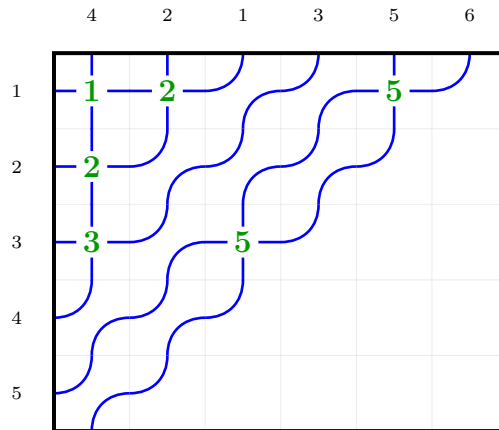
See Figure 1 for an example of this visualization.

The main benefit of this is that visualizing $\text{rt}_R(i, j)$ is easy. The pipes that pass through position (i, j) are labeled s and q for some $s, q > 0$. For a positive root in an unoccupied square, we will have the following labeling, where $q < s$:



We observe that in the above RC graph, the position $(1, 5)$ does not have this configuration. Placing a crossing there would create a negative root, and the pipes would cross twice. *Note that this results in a collection of ordered pairs that is not an RC graph, if such a crossing exists.* For a valid RC graphs, only positive roots occur as crossings, and this happens if and only if pipes cross at most once.

FIGURE 2. An invalid set of crossings causing pipes to cross more than once, caused by inserting a crossing at a negative root



4.3. Zeroing out the last row. We proceed now to define a product on $\mathcal{BRC}$ turning it into a ring. To do this, we need to be able to define a function $\mathcal{Z} : \mathcal{BRC} \rightarrow \mathcal{BRC}$ trimming empty rows from the bottom instead of from the top. This is far more complicated.

See Figure 3 for an example of the zeroing operation (Algorithm 3).

Algorithm 1 Basic monk insertion $i \overset{k}{\rightsquigarrow} R$

-
- 1: **Input:** An RC graph R , parameter k , and an integer i with $1 \leq i \leq k$.
 - 2: **Output:** A modified RC graph R' with $\mathbf{wt}_j(R') = \mathbf{wt}_j(R)$ for $j \neq i$ and $\mathbf{wt}_i(R') = \mathbf{wt}_i(R) + 1$ such that $w_R \xrightarrow{k} w_{R'}$.
 - 3: Find leftmost position $(i, j) \notin R$ in row i such that if $\mathbf{rt}_R(i, j) = (a, b)$, then $a \leq k < b$.
 - 4: $R \leftarrow R \cup \{(i, j)\}$
 - 5: **if** there exists $(i', j') \in R$ with $\mathbf{rt}_R(i', j') \in \Phi^-$ **then**
 - 6: $R \leftarrow R \setminus \{(i', j')\}$
 - 7: Return to step 3 substituting $i = i'$.
 - 8: **end if**
 - 9: **return** R
-

Algorithm 2 Pseudo-Pieri insertion $I \overset{k}{\rightsquigarrow} R$

-
- 1: **Input:** An RC graph R , parameter k , and sequence $I = \{i_1, \dots, i_m\}$ with $k \geq i_1 \geq i_2 \geq \dots \geq i_m \geq 1$.
 - 2: **Output:** A modified RC graph R' with $\mathbf{wt}_i(R') = \mathbf{wt}_i(R)$ for $i \notin I$ and $\mathbf{wt}_i(R') = \mathbf{wt}_i(R) + \#\{j \mid i_j = i\}$ for $i \in I$ such that $w_R \xRightarrow{k} w_{R'}$.
 - 3: **Initialize:** Let $L \leftarrow []$ (empty list of pairs (a, b) where $a \leq k < b$), and $U \leftarrow []$ (empty list of positions).
 - 4: **for** each $i \in (i_1, \dots, i_m)$ **do**
 - 5: Find leftmost position $(i, j) \notin R$ in row i satisfying $j > j'$ for all $(i, j') \in U$ such that either:
 - 6: a) $\mathbf{rt}_R(i, j) = (s, q)$ where $s \leq k < q$ and $q \notin \{b \mid (a, b) \in L\}$.
 - 7: b) $\mathbf{rt}_R(i, j) = (b_r, q)$ where $q > k$, $b_r < q$, and $q \notin \{b \mid (a, b) \in L\}$.
 - 8: $R \leftarrow R \cup \{(i, j)\}$ and $U \leftarrow U \cup \{(i, j)\}$
 - 9: Update L by adding (s, q) or replacing (a_r, b_r) with (a_r, q) and (a_r, b_r) .
 - 10: **if** there exists $(i', j') \in R$ with $\mathbf{rt}_R(i', j') \in \Phi^-$ **then**
 - 11: $R \leftarrow R \setminus \{(i', j')\}$
 - 12: Let $\mathbf{rt}_R(i', j') = (q, s)$
 - 13: **if** $(s, q) \in L$ **then**
 - 14: Remove (s, q) from L
 - 15: **else**
 - 16: Remove (a_r, q) from L , where $(a_r, q), (a_r, s) \in L$
 - 17: **end if**
 - 18: Return to step 5 to insert i' .
 - 19: **end if**
 - 20: **end for**
 - 21: **return** R
-

Note that Algorithm 2 is *not* a realization of Sottile's Pieri formula for complete symmetric polynomials, despite preserving the relevant Bruhat relation. This is because the map

$$\left(\binom{k}{p}\right) \times \mathcal{RC}(w) \rightarrow \mathcal{RC}(n)$$

given by

$$(I, R) \mapsto I \overset{k}{\rightsquigarrow} R$$

need not be injective.

Lemma 4.3.1. *Given an RC graph R , an integer k , and $k \geq i_1 \geq \dots \geq i_m \geq 1$, Algorithm 2 produces a valid RC graph R' satisfying*

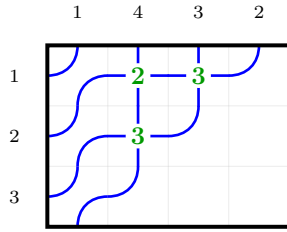
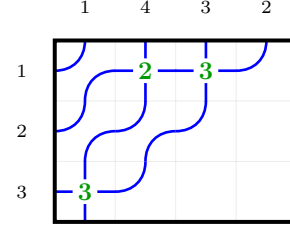
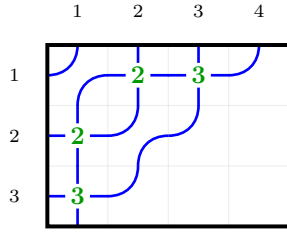
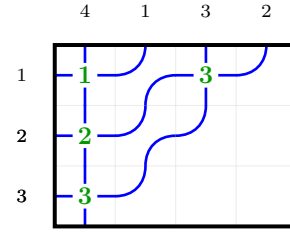
$$\mathbf{wt}_i(R') = \mathbf{wt}_i(R) + \#\{j \mid i_j = i\}$$

and

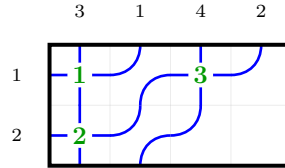
$$w_R \xRightarrow{k} w_{R'}$$

Algorithm 3 Map $\mathcal{Z}(R)$ zeroing out last row

-
- 1: **Input:** A bounded RC graph R with $\text{ht}(R) = n$ and row n empty.
 - 2: **Output:** A bounded RC graph R' with $\text{ht}(R') = n - 1$, $|R'| = |R|$ (same number of crossings), and $w_R \xrightarrow{n} w_{R'}$.
 - 3: **if** R with last row removed is a valid bounded RC graph **then**
 - 4: **return** R with last row removed
 - 5: **else**
 - 6: Let $R_0 \leftarrow R$.
 - 7: Find the maximal p and sequence $\{(i_m, j_m)\}_{m=1}^p$ such that:
 - 8: $\text{rt}_{R_{m-1}}(i_m, j_m) = (n + m - 1, n + m)$ for each m , where $R_m = R_{m-1} \setminus \{(i_m, j_m)\}$ for each $m \geq 1$.
 - 9: $R^- \leftarrow R \setminus \{(i_1, j_1), \dots, (i_p, j_p)\}$.
 - 10: Let $I \leftarrow (i_1, i_2, \dots, i_p)$.
 - 11: $D \leftarrow R^- \cup \{(n, 1), (n, 2), \dots, (n, p)\}$.
 - 12: $D' \leftarrow I \xrightarrow{n-1} D$
 - 13: $R' \leftarrow \{(i, j) \in D' \mid i < n\}$ as a bounded RC graph with $n - 1$ rows.
 - 14: **return** R' .
 - 15: **end if**
-

(a) Initial RC graph R (b) After deleting $(2, 2)$ with $\text{rt}_R(2, 2) = \alpha_3$ and moving to row 3(c) After insertion with $i_1 = 2$ 

(d) After rectification

(e) Final result $\mathcal{Z}(R) = (R', 2)$ after removing row 3FIGURE 3. Step-by-step computation of $\mathcal{Z}(R)$ for $R = \{(1, 2), (1, 3), (2, 2)\}$.

Proof. Set $R_0 = R$ to be the initial RC graph. We claim that given R and L at the start of iteration p of the main loop, we have that R is a valid RC graph satisfying

$$\mathbf{wt}_i(R) = \mathbf{wt}_i(R_0) + \#\{j \leq p-1 \mid i_j = i\}$$

and

$$w_R = w_{R_0} t_{a_1 b_1} \cdots t_{a_{p-1} b_{p-1}}$$

where $L = [(a_1, b_1), \dots, (a_p, b_p)]$. This is clear for $p = 1$. For larger p , suppose we are inserting at row i_p . If we are in case (a) of Step 5, then adding (i_p, j) to R adds the root $t_s - t_q$ to the inversion set of w_R , and multiplying by t_{sq} gives the desired result. If we are in case (b) of Step 5, then adding (i_p, j) to R results in multiplication by $t_{b_r q}$. This commutes past the other reflections to meet $t_{a_r b_r}$. We thus have the situation

$$t_{a_r b_r} t_{b_r q} = t_{a_r q} t_{a_r b_r}$$

This ensures that the product remains as desired if the RC graph is valid. However, the condition that the RC graph R be valid is not guaranteed after this step, so we must rectify. Suppose we must rectify at row $i-1$. We find a negative root $(i-1, j')$ which must have been created by the insertion, and hence by the strong exchange property equal to $\mathbf{rt}_R(i, j)$ that was inserted. Removing this root (from both the graph and from L) returns us to the original permutation at the beginning of the iteration, and performing the insert at row $i-1$ adds a new positive root, giving the desired result on the permutation. The weight is unchanged since we removed and added one crossing in row $i-1$. If the algorithm ends here, we have instead multiplied by the new reflection obtained in the ultimate insert step. Otherwise, repeating this process for all necessary rectifications completes the proof of the claim. The claim iterated to $p = m+1$ yields the desired result, and the method of updating L ensures that the b_j are all distinct, so that $w_R \xrightarrow{k} w_{R'}$. $\square$

Lemma 4.3.2. *Given a bounded RC graph R with $\mathbf{ht}(R) = n$ such that row n is empty, Algorithm 3 produces a valid bounded RC graph R' with $\mathbf{ht}(R') = n-1$ satisfying*

$$\mathbf{wt}(R') = \mathbf{wt}(R)$$

Proof. Stripping out the crossings at roots $(n, n+1), (n+1, n+2), \dots, (n+p-1, n+p)$ removes exactly one crossing from each of rows $i_1, i_2, \dots, i_p$, and moving them to the last row to obtain R_1 ensures that $w_R = w_{R_1}$. By construction, the portion of the RC graph in rows with index less than n is valid (meaning its max descent is at most $n-1$). Applying Algorithm 2 to insert at rows $i_1, i_2, \dots, i_p$ adds different crossings to preserve the number of elements in each row without adding descents past index n in the permutation, by Lemma 4.3.1. Removing row n then yields a valid bounded RC graph R' with $|R'| = n-1$ and the desired properties. $\square$

Theorem 4.3. *Let w be a permutation with last descent at most n . Let $\mathcal{RC}(w, n)^0$ be the set of bounded RC graphs R with $\mathbf{ht}(R) = n$ such that $w_R = w$ and row n is empty. Then*

$$\mathcal{Z} : \mathcal{RC}(w, n)^0 \rightarrow \bigcup_{w \xrightarrow{n} w'} \mathcal{RC}(w', n-1)$$

is a weight-preserving bijection.

To prove Theorem 4.3 for $n = 2$, we may characterize the bounded RC graphs R with $|R| = 2$ such that row 2 is empty and $\mathbf{m}(w_R) = 2$ as those for permutations $w = s_k s_{k-1} \cdots s_2$ for some $k > 1$. Algorithm 3 in this case consists moves the entirety of the crossings in row 1 to row 2. The procedure then inserts $k-1$ crossings into row 1 in order from left to right, which must have roots $t_q - t_s$ such that $q = 1$. Thus R' is precisely $\{(1, 1), (1, 2), (1, 3), \dots, (1, k-1)\}$, which is the unique bounded RC graph for the permutation $w' = s_{k-1} s_{k-2} \cdots s_1$ with one row. This establishes the base case.

For the induction step, we require the following lemmas.

Lemma 4.3.3. *Let R be a bounded RC graph with $\mathbf{ht}(R) = n \geq 2$ such that row n is empty. Then if $\mathcal{Z}(R) = R'$, we have*

$$w_R \xrightarrow{n} w_{R'}$$

Proof. Let $w = w_R$ and suppose $\mathfrak{c}_n(w) = m$. By construction, the second to last step of Algorithm 3 yields a bounded RC graph $\tilde{R}$ with $|\tilde{R}| = n$ such that if $\tilde{w} = w_{\tilde{R}}$, then

$$\tilde{w}(n) > \tilde{w}(n+1) < \tilde{w}(n+2) < \cdots < \tilde{w}(n+m)$$

and

$$w \xrightarrow{n-1} \tilde{w}$$

via reflections t_{ab} such that $n \leq b \leq n+m$. Let $w' = w_{R'}$. We also have

$$\tilde{w} \xrightarrow{n} \varphi_{n,N}(w')$$

via only reflections t_{nb} such that $b > n+m$. We claim that

$$w \xrightarrow{n} \varphi_{n,N}(w')$$

This can only fail if $\tilde{w}(n) \neq w(n)$ by the observations above. However, the pipe labeled n , by virtue of the fact that we have $(n, 1), \dots, (n, m)$ populated, will always lie to the right of pipes $n+1, n+2, \dots, n+m$ in the wiring diagram for $\tilde{w}$ or any intermediate stage. Inserting into any row that contained a crossing $(n, n+p)$, there must be a valid empty space to the left of the pipe labeled n since we have deleted these crossings. By virtue of the fact that the leftmost is always chosen, no reflection (i, n) will ever be inserted. This ensures that $\tilde{w}(n) = w(n)$, as desired. $\square$

Lemma 4.3.4. *Let (R, n) be a bounded RC graph with $n \geq 2$ such that row n is empty. Then*

$$\mathcal{Z}(\text{trim}(R)) = \text{trim}(\mathcal{Z}(R))$$

Proof. This is almost a trivial observation. In terms of the word of R , $\text{trim}(R, n)$ preserves a suffix. Removing all of the initial roots before trimming is therefore the same as removing the initial roots that still remain after trimming, by the exchange property. Afterwards, in performing the insertion algorithm, there is no dependency on rows with a lower index, the modification only proceeds downward in row number. Therefore, trimming the first row at any point only stops the process earlier, and does not change rows with higher index than 1 in the outcome. $\square$

Lemma 4.3.5. *Let w be a permutation with last descent at most n . Let $\mathcal{RC}(w, n)^0$ be the set of bounded RC graphs R with $\text{ht}(R) = n$ such that $w_R = w$ and row n is empty. Then*

$$\mathcal{Z} : \mathcal{RC}(w, n)^0 \rightarrow \mathcal{RC}(n-1)$$

is injective.

Proof. If $n = 2$ this is clear. Suppose now that $n > 2$ and that the result holds for $n-1$. Let $(R, n) \in \mathcal{RC}(w, n)^0$. If $(R', n) \in \mathcal{RC}(w, n)^0$ is another bounded RC graph such that

$$\mathcal{Z}(R) = \mathcal{Z}(R')$$

then by Lemma 4.3.4, we have

$$\mathcal{Z}(\text{trim}(R)) = \mathcal{Z}(\text{trim}(R'))$$

hence R and R' agree on all rows except possibly row 1 by the inductive hypothesis. Since $w_R = w_{R'}$, it follows that $R = R'$, establishing injectivity. $\square$

Proof of Theorem 4.3. By the previous lemma, it suffices to show that $\mathcal{Z}$ is surjective. However, this follows from the pigeonhole principle and the well-known transition formula for Schubert polynomials. $\square$

We may extend $\mathcal{Z}$ to an endomorphism $\mathcal{BRC} \rightarrow \mathcal{BRC}$ by defining $\mathcal{Z}(R) = 0$ if the last row of R is not empty. Then we have the following result.

Theorem 4.4. *Let w be a permutation with last descent at most n . Then*

$$\mathcal{Z}(\mathcal{S}_w(n)) = \sum_{\substack{w \xrightarrow{n} w' \\ \ell(w) = \ell(w')}} \mathcal{S}_{w'}(n-1)$$

4.4. Preservation of the Demazure crystal structure. RC-graphs have a Demazure crystal structure defined by Assaf and Schilling in [1]. Namely, they define operators $e_i : \mathcal{RC}(w) \rightarrow \mathcal{RC}(w) \cup \{\emptyset\}$ and $f_i : \mathcal{RC}(w) \rightarrow \mathcal{RC}(w) \cup \{\emptyset\}$ for each $i \geq 1$ as follows.

Definition 4.4.1. Consider elements $(i, j) \in R$ and $(i + 1, k) \in R$. The pairing algorithm starts with the largest j in row i and pairs (i, j) with $(i + 1, k)$ where k is the smallest such that $k \geq j$. If no such k exists, then (i, j) is unpaired. The algorithm continues by considering the next largest j in row i and repeating the process until all elements in row i have been considered.

Define R_i to be the set of unpaired elements in row i after applying the pairing algorithm between rows i and $i + 1$, and define L_i to be the set of unpaired elements in row $i + 1$. Then $f_i(R)$ is defined by removing the leftmost element (i, j) from R_i and adding the element $(i + 1, k)$ to R , where k is the largest value such that $(i + 1, k) \notin R$ and $k < j$. If R_i is empty or k does not exist, then $f_i(R) = \emptyset$. Similarly, $e_i(R)$ is defined by removing the rightmost element $(i + 1, k)$ from L_i and adding the element (i, j) to R , where j is the smallest value such that $(i, j) \notin R$ and $j > k$. If L_i is empty, then $e_i(R) = \emptyset$.

We may transport this structure to bounded RC graphs as operators $e_i : \mathcal{RC}(w, n) \rightarrow \mathcal{RC}(w, n) \cup \{\emptyset\}$ and $f_i : \mathcal{RC}(w, n) \rightarrow \mathcal{RC}(w, n) \cup \{\emptyset\}$ by preserving the bound:

$$e_i(R) = e_i(R)$$

$$f_i(R) = f_i(R)$$

where the bound is preserved (i.e., $\text{ht}(f_i(R)) = \text{ht}(R)$ when $f_i(R) \neq \emptyset$).

Definition 4.4.2. An RC graph R has a uniquely associated reduced word $\text{word}(R) = (s_{i_1}, \dots, s_{i_N})$. Recall the Coxeter-Knuth relation $\tilde{\text{eg}}$ defined by

$$j \ i \ k \sim j \ k \ i$$

and

$$k \ i \ j \sim i \ k \ j$$

if $i < j < k$, and

$$i \ i + 1 \ i \sim i + 1 \ i \ i + 1$$

For an RC graph R , define $N(R)$ to be the maximum letter in R . Then define

$$\tilde{\text{eg}}(R) = (N(R) + 1 - i_1, \dots, N(R) + 1 - i_N)$$

Via the Edelman-Greene insertion algorithm, we may define a pair

$$(P, Q)$$

of tableaux of the same shape such that the reading word of P is CK-equivalent to $\tilde{\text{eg}}(R)$ and Q , the recording tableau, is standard. Define $\text{tab}(R)$ to be the tableau of the same shape as Q such that the entry in box (i, j) is the row index of the box in R that was inserted to create box (i, j) in P .

Lemma 4.4.3. *We have that $\text{tab}(R)$ is a semistandard Young tableau of shape $\text{shape}(Q)$, and the map*

$$R \mapsto \text{tab}(R)$$

is a morphism of crystal graphs into $B(\text{shape}(Q))$.

Proof. Suppose R is a highest weight. Then $\text{wt}(R)$ is a partition. By the highest weight condition, the longest increasing subsequence of $\tilde{\text{eg}}(R)$ is of length $\text{wt}(R)_1$ and is precisely the first row read in the grid order. By the Edelman-Greene correspondence [5], the first row of P has length equal to the length of the longest increasing subsequence of $\tilde{\text{eg}}(R)$. Hence, the first row of P has length $\text{wt}(R)_1$. By similar reasoning on the subsequent rows, we have that $\text{shape}(P) = \text{wt}(R)$. We then have that $\text{tab}(R)$ is the unique Yamanouchi tableau of shape $\text{wt}(R)$, establishing the claim for highest weights. The general case follows from the rigidity of the Demazure crystal structure [1]; since the function is weight-preserving, it is a morphism of crystal graphs. $\square$

Theorem 4.5. *The distinct Demazure crystals that make up $\mathcal{RC}(w)$ are in bijection with the Coxeter-Knuth equivalence classes of reduced words for $w_0 w w_0$ via the map $R \mapsto \tilde{\text{eg}}(R)$, with $R \mapsto \text{tab}(R)$ being a homomorphism of crystal graphs into $B(\lambda)$, where $\lambda = \text{shape}(P(\tilde{\text{eg}}(R)))$.*

Proof. By [9, Theorem 4.11], if $e_i(R) \neq \emptyset$, then R and $e_i(R)$ have the same Edelman-Greene insertion tableau. Since conjugation by w_0 is an automorphism of Coxeter-Knuth equivalence, so do $\widetilde{\text{eg}}(R)$ and $\widetilde{\text{eg}}(e_i(R))$. The shape is determined inductively. Suppose R is a highest weight element and consider $\text{word}(R)$. The first row of R forms a word that corresponds to an initial increasing subsequence of $\widetilde{\text{eg}}(R)$. If there were a letter in some row below the first that is smaller than the last letter of the first row, then some raising operator would yield a nonempty result, contradicting the fact that R is a highest weight. Thus, the first row of $P(\widetilde{\text{eg}}(R))$ has length equal to the length of the first row of R . The rest follows by induction on the remaining rows. $\square$

Definition 4.4.4. Let $\mathcal{RC}(w, n)^0$ be the set of bounded RC graphs R with $\text{ht}(R) = n$ such that $w_R = w$ and row n is empty. We define a function $\mathcal{Z}_*^R : R \rightarrow R'$, where $\mathcal{Z}(R) = (R', n-1)$ and $w' = w_{R'}$, as follows. Clearly if $\mathfrak{c}_n(w) = 0$, then $\mathcal{Z}_*^R$ is the identity map.

Consider now the case where $\mathfrak{c}_n(w) = 1$. The initial deletion of (i, j) with $\text{rt}_R(i, j) = (n, n+1)$ results in an RC graph $R_0 = R \setminus \{(i, j)\}$ with $w_{R_0} = w_{s_n}$, after which we add $(n, 1)$ to obtain R_1 . Let

$$f_1 : R_1 \rightarrow R$$

be the identity. Afterwards, (i_1, j_1) is inserted into R_1 to obtain R_2 . Set

$$f_1(i_1, j_1) = (i, j)$$

and

$$f_1(a, b) = (a, b)$$

otherwise.

At this point, rectification may be required. In that case, we delete some element (i'_1, j'_1) from R_2 such that $i'_1 < i_1$ and $\text{rt}_{R_2}(i'_1, j'_1) = \text{rt}_{R_2}(i_1, j_1)$. Afterwards, we add (i'_1, j_2) to obtain R'_2 . Set

$$f_2(i'_1, j_2) = (i'_1, j'_1)$$

and

$$f_2(a, b) = f_1(a, b)$$

otherwise. Continuing in this manner, we eventually obtain R' . Piecing all of the maps defined at each step gives the desired map. The same method works for $\mathfrak{c}_n(w) > 1$ by induction.

Lemma 4.4.5. Let $(R, n) \in \mathcal{RC}(w, n)^0$ and suppose $\mathcal{Z}(R) = (R', n-1)$. Then for any $(a, b) \in R'$, (a, b) is paired in the $i, i+1$ pairing in R' if and only if $\mathcal{Z}_*^R(a, b)$ is paired in the $i, i+1$ pairing in R .

Proof. This is true for $a = i$ because at most one root is changed in row i and it has moved to the left. For $a = i+1$, the only way that the pairing status could change is if a root in row i moved past an unpaired root in row $i+1$. However, by construction of the insertion algorithm, this cannot happen since a root that was already inserted into row $i+1$ was deleted from row i in the rectification process, and the new inserted root occurs to the left. $\square$

Theorem 4.6. Let $w \in S_\infty$ and $n > 0$ be such that $\mathfrak{m}(w) \leq n$. Then

$$\mathcal{Z} : \mathcal{RC}(w, n)^0 \rightarrow \bigcup_{w \searrow_n w'} \mathcal{RC}(w', n-1)$$

is a weight-preserving isomorphism of crystal graphs.

Proof. The existence of the map $\mathcal{Z}_*^R$ and the previous lemma establish that $\mathcal{Z}$ is a bijection (by Theorem 4.3) that preserves both weights and the lengths of i -strings. Hence it is a weight-preserving isomorphism of crystal graphs. $\square$

4.5. Definition of the ring product.

Definition 4.5.1. Suppose we have two bounded RC graphs R_1 with $\text{ht}(R_1) = m$ and R_2 with $\text{ht}(R_2) = n$. The product of these is defined by

$$R_1 \diamond R_2 = \sum_{\substack{\text{clip}^m(R')=R_1 \\ \text{trim}^m(R')=R_2}} R'$$

where R' ranges over bounded RC graphs with $\text{ht}(R') = m+n$.

Define a map $D : \mathcal{BRC} \rightarrow \mathcal{D} \otimes \mathcal{D}$ by

$$D(R) = \Xi_{w_R}^{\text{ht}(R)} \otimes [\text{wt}(R)]$$

Theorem 4.7. *The product $\diamond$ turns $\mathcal{BRC}$ into a ring, and $D : \mathcal{BRC} \rightarrow \mathcal{D} \otimes \mathcal{D}$ is a homomorphism of rings.*

Proof. What is in question is associativity. Suppose we have three bounded RC graphs R_1 with $\text{ht}(R_1) = p$, R_2 with $\text{ht}(R_2) = q$, and R_3 with $\text{ht}(R_3) = r$, and we want to show that the order of bracketing the product is irrelevant. This is because if R_w is a term in the expansion of the product with $\text{ht}(R_w) = p + q + r$, then

$$\begin{aligned} \text{clip}^p(\text{clip}^{p+q}(R_w)) &= R_1 = \text{clip}^p(R_w) \\ \text{trim}^p(\text{clip}^{p+q}(R_w)) &= R_2 = \text{clip}^q(\text{trim}^p(R_w)) \end{aligned}$$

and

$$\text{trim}^{p+q}(R_w) = R_3 = \text{trim}^q(\text{trim}^p(R_w))$$

by Lemma 4.5.2, and each term in the expansion of the product has a coefficient of 1.

To prove the map $D : \mathcal{BRC} \rightarrow \mathcal{D} \otimes \mathcal{D}$ is a homomorphism of rings, we utilize the formula for the coefficient $d_{u,v}^w(p, q)$ as the coefficient of $\mathfrak{S}_u(x_1, \dots, x_p)$ when we set $x_{p+1} = \dots = x_{p+q} = 0$ in the polynomial

$$\partial^{\uparrow p v} \mathfrak{S}_w(x_1, \dots, x_{p+q})$$

(that is, Theorem ??). We claim that if we fix an RC graph R_u for u and an RC graph R_v for v , then this coefficient is the same as the number of BRCs $(R, p+q)$ for w such that $\text{trim}^p(R) = R_v$ and $\text{clip}^p(R) = R_u$. If we assume that v is the identity, this follows from Theorem 4.4 and Theorem 4.3. The divided difference operator characterization means that for $\ell(v) > 0$, we can replace the empty space with any (shifted up) RC graph for v (R_v, q) and get the result for α as desired. Since the weight is also preserved in this construction, the result follows for D as well. $\square$

Lemma 4.5.2. *We have the following equations for an RC graph R and two integers p and q :*

$$\begin{aligned} \text{clip}^p(\text{clip}^{p+q}(R)) &= \text{clip}^p(R) \\ \text{trim}^p(\text{clip}^{p+q}(R)) &= \text{clip}^q(\text{trim}^p(R)) \end{aligned}$$

and

$$\text{trim}^{p+q}(R) = \text{trim}^q(\text{trim}^p(R))$$

Proof. Suppose (R, n) is a bounded RC graph. We prove that $\text{clip}^{n-2}(\text{clip}^{n-1}(R)) = \text{clip}^{n-2}(R)$. For $n = 2$, this is trivial. As an additional base case, we also need $n = 3$, which is not trivial, but true because a bounded RC graph $(R, 1)$ is uniquely determined by its size. We proceed by induction on n . Without loss of generality, assume that row n is empty. Then $\text{clip}^{n-1}(R) = \mathcal{Z}(R)$. We have

$$\begin{aligned} \text{clip}^{n-2}(R) &= \mathcal{Z}^2(R_{\leq n-2}, n) = \mathcal{Z}(\mathcal{Z}(R_{\leq n-2})) \\ \text{clip}^{n-2}(\text{clip}^{n-1}(R)) &= \mathcal{Z}(\mathcal{Z}(R)_{\leq n-2}, n-1) \end{aligned}$$

IF $\mathfrak{m}(w_{R_{\leq n-2}}) \leq n-1$, then both sides are equal to $\mathcal{Z}(R_{\leq n-2})$. Otherwise, by Lemma 4.3.4, we have

$$\begin{aligned} \text{trim}(\text{clip}^{n-2}(R)) &= \text{trim}(\mathcal{Z}(\mathcal{Z}(R_{\leq n-2}))) = \mathcal{Z}(\mathcal{Z}(\text{trim}(R)_{\leq n-3})) \\ \text{trim}(\text{clip}^{n-2}(\text{clip}^{n-1}(R))) &= \mathcal{Z}(\mathcal{Z}(\text{trim}(R))_{\leq n-3}, n-2) \end{aligned}$$

Thus, $\text{clip}^{n-2}(\text{clip}^{n-1}(R))$ and $\text{clip}^{n-2}(R)$ agree on all rows except possibly the first row. We proceed by induction on the number of elements of R in the first row. For zero elements, the result is obvious. If we have proved the result for $k-1$ elements in the first row, let $(1, i)$ be the rightmost such element. Then $R \setminus \{(1, i)\}$ has $k-1$ elements in the first row, so by the inductive hypothesis, $\text{clip}^{n-2}(\text{clip}^{n-1}(R \setminus \{(1, i)\})) = \text{clip}^{n-2}(R \setminus \{(1, i)\})$. Applying Theorem ??, we see that adding back in the element $(1, i)$ to both sides preserves equality, due to the commutativity of the diagram. This establishes the first equation. The second equation easily follows from repeated application of Lemma 4.3.4, and the third equation is the definition of trim . $\square$

Theorem 4.8. *Let $r > 0$ be an integer and suppose $p \geq 0$ is an integer such that $p \leq r$. Then the map $\mathcal{BRC}(r) \rightarrow \mathcal{BRC}(p) \times \mathcal{BRC}(r-p) \times S_\infty$ defined by*

$$(R, r) \mapsto (\text{clip}^p(R, r), \text{trim}^p(R, r), w_R)$$

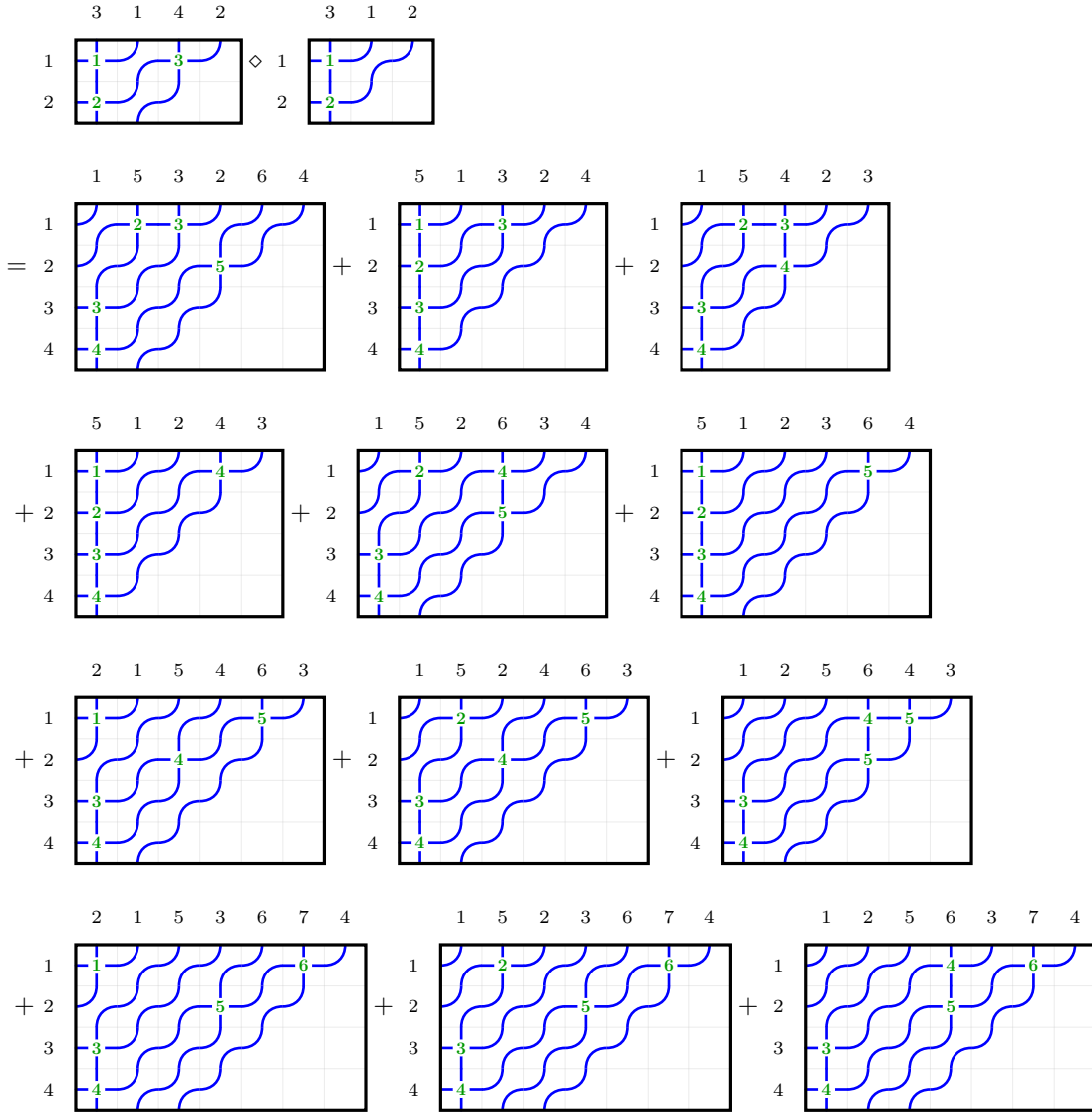
is injective.

Proof. For $p = 0$ this is trivial, and for $p = 1$ this is due to the fact that any subset of the first row of an RC graph corresponds to a unique permutation, so that if we know $\text{trim}^1(R, r) = (R', r - 1)$ and we know w_R , then R is the unique RC graph such that the first row corresponds to the permutation u such that $uw_{R'} = w_R$ and $\text{trim}^1(R, r) = (R', r - 1)$.

For $p = r - 1$, we use the fact that row r of a bounded RC graph (R, r) is uniquely determined by its weight. It therefore suffices to prove the result when this weight is 0, which is covered by Theorem 4.3. We have thus proved the theorem for $r = 2$.

For larger r , we prove the result by induction on p , keeping in mind the solutions for the cases $p = 0, 1, r - 1$. We know that (R, r) is uniquely determined by $(\text{clip}^{r-1}(R), w_R)$ and we may use the inductive hypothesis to reason that it is therefore uniquely determined by $(\text{clip}^p(R), \text{trim}^p(\text{clip}^{r-1}(R)), w_R)$. Furthermore, Let $u = w_{\text{clip}^p(R)}$. Then the pair $(\text{trim}^{p-1}(\text{clip}^{r-1}(R)), u^{-1}w_R)$ uniquely determines $\text{trim}^p(R)$ by the inductive hypothesis. Hence $(\text{clip}^p(R), \text{trim}^p(R), w_R)$ uniquely determines (R, r) , establishing injectivity. $\square$

Example 4.5.3. We compute the product of two bounded RC graphs $(R_1, 2) \in \mathcal{BRC}(2413, 2)$ and $(R_2, 2) \in \mathcal{BRC}(231, 2)$:



Definition 4.5.4. Let $\mathcal{R}$ be the subring generated by the single-row RC graphs $\text{row}(i)$ for $i \geq 0$. Let a be a sequence of elements of a ring indexed by the nonnegative integers. For a permutation w and an integer

$n \geq \mathbf{m}(w)$, define a function $\Xi_w^n(a)$ By

$$\Xi_w^n(a) = \sum_{\alpha} d_{\alpha,w}^n a_{\alpha_1} \cdots a_{\alpha_n}$$

where the sum is over all weak compositions α of length n , and $d_{\alpha,w}^n$ is the coefficient of α in $\Xi_w^n \in \mathcal{D}$.

Proposition 4.5.5. *We have that the elements*

$$\Xi_w^n(\mathbf{row})$$

as n ranges over all nonnegative integers w ranges over all permutations in S_{∞} with $n \geq \mathbf{m}(w)$ form a $\mathbb{Z}$ -basis of $\mathcal{R}$, and $\alpha : \mathcal{R} \rightarrow \mathcal{D}$ is an isomorphism of graded rings.

Proof. Since $\alpha(\Xi_w(\mathbf{row})) = \Xi_w^n \in \mathcal{D}$, the elements $\Xi_w^n(\mathbf{row})$ are linearly independent. Given that the images form a basis of $\mathcal{D}$, they must also span $\mathcal{R}$ since $\mathcal{D}$ is a free algebra, and hence there is a homomorphism backwards $\mathcal{D} \rightarrow \mathcal{R}$ sending $[i] \mapsto \mathbf{row}(i)$ which is necessarily surjective. Thus, the elements $\Xi_w^n(\mathbf{row})$ form a $\mathbb{Z}$ -basis of $\mathcal{R}$. $\square$

4.6. Relation to Little bumps. Define a function $\mathcal{L} : \mathcal{BRC}^0(n) \rightarrow \mathcal{BRC}(n+1)$ as follows. Assume $n = \mathbf{m}(w_R)$, and suppose $\mathbf{rt}_R(i, j) = (n, n+1)$, where $i < n$. Set

$$R' = (R \setminus \{(i, j)\}) \cup \{(n, 1)\}$$

Then define

$$\mathcal{L}(R, n) = ((i \xrightarrow{n-1} R') \setminus \{(n, 1)\}, n+1)$$

Then define $\mathcal{Z} : \mathcal{BRC}^0(n) \rightarrow \mathcal{BRC}(n-1)$ by iterating $\mathcal{L}$ and stopping as soon as the result has $\mathbf{m}(w_R) \leq n-1$.

Proposition 4.6.1. *Let $(R, n) \in \mathcal{BRC}^0(n)$ be a bounded RC graph such that $\mathbf{m}(w_R) = n$. Suppose*

$$\mathbf{word}(R) = a_1 a_2 \cdots a_k$$

and

$$\mathbf{seq}(R) = r_1 r_2 \cdots r_k$$

suppose $\mathbf{rt}_R(i, j) = (n, n+1)$, and let p be the index such that $r_p = i$ and $r_p + j - 1 = a_p$. Then

$$\mathbf{word}(\mathcal{L}(R, n)) = \mathbf{little}_p(\mathbf{word}(R, n))$$

Proof. Consider the biword

$$\begin{array}{cccccccc} r_1 & r_2 & \cdots & r_{p-1} & r_p & r_{p+1} & \cdots & r_k \\ a_1 & a_2 & \cdots & a_{p-1} & a_p & a_{p+1} & \cdots & a_k \end{array}$$

The first deletion yields

$$\begin{array}{cccccccc} r_1 & r_2 & \cdots & r_{p-1} & r_{p+1} & \cdots & r_k & n \\ a_1 & a_2 & \cdots & a_{p-1} & a_{p+1} & \cdots & a_k & n \end{array}$$

Necessarily, we must have $a_p > 1$. If we have $a_p = 1$, then we would have $r_p = 1$. The only way for position $(1, 1)$ in an RC graph to be a right descent is if it is the unique right descent. By assumption, that descent would have to be n , so that $n = 1$. In that case, however, it would not be possible for the last row to be empty.

We also must have that $r_p < a_p$, because there is a sequence of chute moves involving this crossing that moves it to position $(n, 1)$ by [2, Theorem 3.7]. Assume first that $p = k$. Then the insertion algorithm will insert $a_p - 1 = n - 1$ into row r_k . If this word is reduced, then we are done, and this clearly coincides with the Little bump. If it is not reduced, the descent $n - 1$ occurs somewhere to the left, say at position j . The insertion algorithm would delete this letter then try to insert another one. However, we can see that recursively this is the same as applying $\mathcal{Z}_+$ to the biword obtained by removing all letters after position j and then reinserting them afterwards. By the inductive hypothesis, this coincides with applying the Little bump at position j , which would indeed be the next step in the recursion of applying the Little bump at position k . Thus, the two coincide by induction. $\square$

Theorem 4.9 ([7, Theorem 4.4]). *Let w be a reduced word for a permutation $w \in S_{\infty}$. Then*

$$Q(w) = Q(\mathbf{little}_p(w))$$

for any valid p .

Corollary 4.6.2. $\mathcal{L} : \mathcal{BRC}^0(n) \rightarrow \mathcal{BRC}(n+1)$ is a weight-preserving injection that preserves the recording tableau under the Edelman-Greene correspondence.

Corollary 4.6.3.

$$\mathcal{L} : \mathcal{RC}(w, n)^0 \rightarrow \bigcup_{w \xrightarrow{n} w'} \mathcal{RC}(w', n-1)$$

is a weight-preserving isomorphism of crystal graphs.

Corollary 4.6.4. We have the equality

$$\mathcal{L} = \mathcal{Z}$$

Proposition 4.6.5. The map $\alpha \times \omega : \mathcal{BRC} \rightarrow \mathcal{D} \times \mathcal{D}$ is a homomorphism of rings, and on the subalgebra $\mathcal{D}$ generated by the single-row RC graphs, it is an isomorphism of rings onto the diagonal.

Proposition 4.6.6. Let $w \in S_\infty$ and let $A_{w,a}$ be the coefficients such that

$$\Xi_w^n = \sum_a A_{w,a} [a]$$

and let $F_{a,w}$ be the coefficients such that

$$[a] = \sum_w F_{a,w} \Xi_w^n$$

Then

then

$$\begin{aligned} \sum_{S \in \mathcal{RC}_n} A_{w, \mathbf{wt}(S)} \omega(S) &= \Xi_w^n \\ \sum_{R, S \in \mathcal{RC}_n} A_{w, \mathbf{wt}(S)} F_{\mathbf{wt}(S), w_R} \alpha(R) &= \Xi_w^n \end{aligned}$$

We have that

$$\Delta(\Xi_w^n) = \sum_{S \in \mathcal{RC}_n} A_{w, \mathbf{wt}(S)} \Delta(\omega(S))$$

This is equal to

$$\Delta(\Xi_w^n) = \sum_{S \in \mathcal{RC}_n, S_1 \in \mathcal{RC}(a), S_2 \in \mathcal{RC}(b), a+b=\mathbf{wt}(S)} A_{w, \mathbf{wt}(S)} \omega(S_1) \otimes \omega(S_2)$$

Change basis to α ,

$$\Delta(\Xi_w^n) = \sum_{S \in \mathcal{RC}_n, S_1 \in \mathcal{RC}(a), S_2 \in \mathcal{RC}(b), a+b=\mathbf{wt}(S)} A_{w, \mathbf{wt}(S)} F_{\mathbf{wt}(S_1), R_1} F_{\mathbf{wt}(S_2), R_2} \alpha(R_1) \otimes \alpha(R_2)$$

4.7. The space of noncommutative elementary monomials.

4.8. Relation to the bijection with bumpless pipe dreams.

Definition 4.8.1. Denote by $\mathcal{BPD}(w, n)$ the set of (reduced) n -row bumpless pipe dreams for a permutation w , where $n \geq \mathbf{m}(w)$. These are $n \times N$ grids, where N is at least the length of w , populated with 6 possible tiles, shown in Figure 4, satisfying the conditions that pipes must enter from the bottom and exit on the right, and no two pipes may cross twice. We may assign a permutation to such a BPD as follows.

Let (B, n) be an n -row BPD. Let $c_1, \dots, c_m$ be the columns such that the pipes on the bottom edge have an entrance from the bottom. If the pipe coming from c_i exits on row j on the right, then $w(c_i) = j$. To fill in the remaining elements of the permutation, we set $w(m+k) = d_k$, where $d_1, \dots, d_k$ are the remaining columns with no entrance in increasing order.

Lemma 4.8.2 ([14, Theorem 3.6]). Let (B, N) be an ordinary reduced BPD for a permutation w , so that $w(M) = M$ for all $M > N$ and B is $N \times N$. Let $n \geq \mathbf{m}(w)$ be an integer. Then the BPD B' obtained by only retaining the first n rows of B satisfies $w(B', n) = w(B, N)$ and uniquely determines B under this assumption.

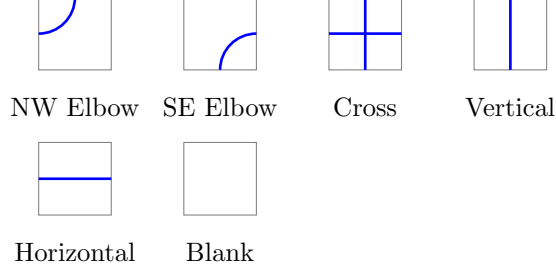


FIGURE 4. The six possible tiles in bumpless pipe dreams. Pipes enter from the bottom and exit at the right.

Definition 4.8.3. There is a bijection $\varphi : \mathcal{BPD}(w, n) \rightarrow \mathcal{RC}(w, n)$, where $w \in S_n$, defined in [6] that preserves weight. It is viewed as “canonical” because it also provides bijective correspondences between certain insertion/bumping operations on BPDs and RC graphs, showing that they agree in a strong way. In light of Lemma 4.8.2, we may extend this bijection to $w \in S_\infty$ with $\mathbf{m}(w) \leq n$ by first expanding the BPD to a square BPD with sufficiently many rows and columns, then applying the bijection, and finally truncating the resulting RC graph to n rows.

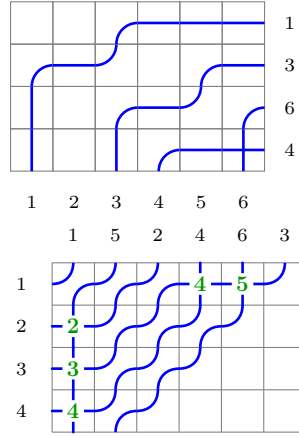


FIGURE 5. An example of a 4-row bumpless pipe dream for a permutation of length 6. Pipes enter from the bottom at columns 1, 3, 4, and 6, and exit on the right at the labeled rows. The associated permutation is $[1, 3, 6, 4, 2, 5]$. The associated RC graph under the bijection φ is shown as well, with crosses in positions $(1, 4)$, $(1, 5)$, $(2, 1)$, $(3, 1)$, and $(4, 1)$.

Definition 4.8.4. We define a function $\mathbf{t} : \mathcal{BPD}(n)^0 \rightarrow \mathcal{BPD}(n-1)$, found in [13] in a more general context, as follows. Given a bumpless pipe dream (B, n) with no blank tiles in row n , we simply remove row n to obtain B' . Then

$$\mathbf{t}(B, n) = (B', n-1)$$

One may ask oneself here whether anything changes at all. Indeed, all that changes are the labels of the incoming pipes, as shown in Figure 6, and this only happens if $\mathbf{m}(w) = n$.

Theorem 4.10. *We have the formula*

$$\varphi \circ \mathbf{t} = \mathcal{Z} \circ \varphi$$

Proof. Little bumps on last descent □

Definition 4.8.5. Let (B, p) be a BPD and let (C, q) be another BPD. We define an operation $B \bowtie C$ as follows. First, let $\varphi(B) = ((a_1, r_1), (a_2, r_2), \dots, (a_m, r_m))$. Then define

$$B \bowtie^p C = \Delta(((a_1, r_1), (a_2, r_2), \dots, (a_m, r_m)), C \uparrow^p)$$

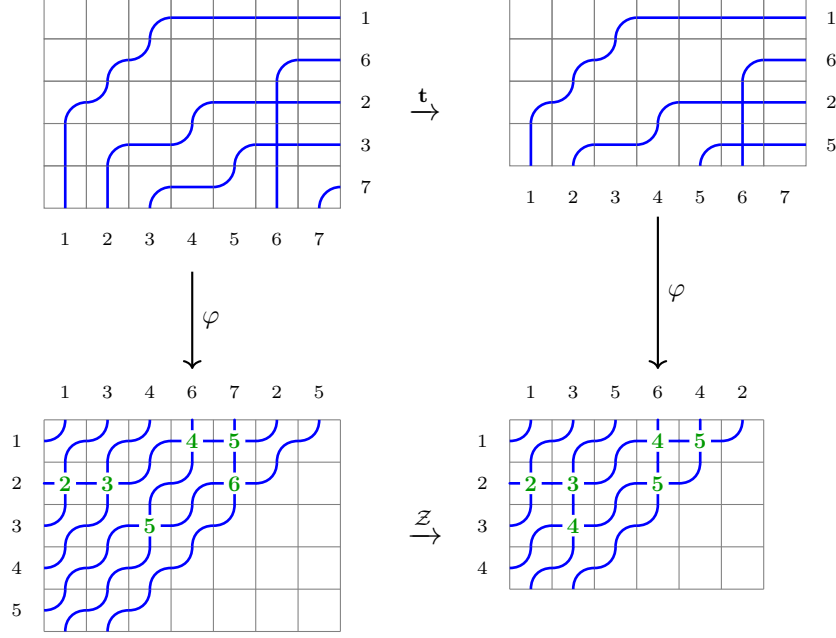


FIGURE 6. The $\mathbf{t}$ transformation on a bumpless pipe dream with permutation $[1, 6, 2, 3, 7, 4, 5]$. The bottom row is removed, resulting in a BPD with one fewer row and permutation $[1, 6, 2, 5, 3, 4]$. The associated RC graphs under the bijection are shown below with the corresponding $\mathcal{Z}$ transformation.

We may define a ring product on $\mathcal{BPD}$, somewhat independently of the bijection, by declaring that

$$(B, p) \diamond (C, q) = \sum_{\mathbf{t}^N(B', p+N) = (B, n)} (B' \curvearrowright^p C, p+q)$$

Squash prod BPDs

Theorem 4.11. *The Gao-Huang bijection $\varphi : \mathcal{BPD} \rightarrow \mathcal{BRC}$ is an isomorphism of rings.*

5. COMPLEXITY OF THE TRANSITION MULTIPLICATION ALGORITHM

In this section we analyze the worst-case running time of `schubmult.py`, the transition-equation algorithm for multiplying ordinary Schubert polynomials, and compare it against dense polynomial multiplication. The bounds we obtain are output-sensitive in a strong sense: they depend on the *total multiplicity*

$$M_{\text{out}}(u, v) := \sum_w c_{u,v}^w$$

of the product $\mathfrak{S}_u \cdot \mathfrak{S}_v$ rather than on $n!$, with cancellation overhead controlled by explicit Bruhat-fiber data. Note that M_{out} dominates the support size $N_{\text{out}}(u, v) := \#\{w : c_{u,v}^w \neq 0\}$ but can be much larger when the structure constants $c_{u,v}^w$ exceed 1.

Throughout, fix $u, v \in S_n$ and write $\theta = \theta(v^{-1})$ for the sequence of column lengths used in the algorithm, with $r = \ell(\theta)$ parts and $\sum_j \theta_j = \ell(v)$. At each level j , the algorithm maintains a dictionary `vpathsumsj` : $(u', v') \mapsto \text{coeff}$, and the “descent operators” applied through `kdown_perms` are the only source of negative signs.

5.1. Baseline rigorous bound. Combining the multiplicity-free Pieri fan-out $\binom{k}{p} \binom{n+1-k}{p}$ with the Chu–Vandermonde identity, the intermediate sets satisfy the unconditional bound

$$|\text{vpathsums}_j| \leq \prod_{i \leq j} \binom{n+1}{\theta_i},$$

and the entire algorithm runs in time

$$(5.1.1) \quad T_{\text{schub}}(u, v) \leq \text{poly}(n) \cdot \prod_{j=1}^r \binom{n+1}{\theta_j}.$$

5.2. The dominant case: no cancellation.

Proposition 5.2.1. *If v is dominant, then every coefficient computed by `schubmult.py` during the run on (u, v) is non-negative, and no two contributions to the same `vpathsums` entry have opposite signs.*

Proof. When v is dominant, v^{-1} is dominant, so $\mu := \text{uncode}(\theta)$ equals v^{-1} and $v\mu = \text{id}$. At every level j , the auxiliary `monopermj` is itself dominant, and the only vp in `vpathdictsj` that survives is reached by the trivial branch `(perm, 0, +1)` of `kdown_perms`. Inspection of `kdown_perms` shows that negative signs are produced only when a non-trivial Bruhat descent of `monopermj` is taken (the line $s_2 := -s$); since no such descent contributes here, every entry of `vpathsums` is computed by an unsigned sum of positive contributions. $\square$

Corollary 5.2.2. *If v is dominant,*

$$T_{\text{schub}}(u, v) = O(M_{\text{out}}(u, v) \cdot \text{poly}(n)),$$

i.e. the algorithm is output-sensitive in the total-multiplicity sense.

Proof. By Proposition 5.2.1 no entry of any `vpathsumsj` can be cancelled to zero in a later step, so every weighted contribution that the algorithm processes survives to the final output. The total work is then proportional to the sum, over all paths through the Pieri tree, of the absolute value of the contribution carried; in the cancellation-free dominant case this is precisely $M_{\text{out}}(u, v)$ up to a polynomial factor in n for the per-step Pieri fan and dictionary operations. Note that intermediate `vpathsums` keys may have multiplicity > 1 , which is correctly counted by M_{out} but not by the bare support size N_{out} . $\square$

5.3. The general case: cancellation is fiber-confined. For non-dominant v , signed contributions can in principle cancel. The following proposition isolates exactly where cancellation can occur.

Proposition 5.3.1. *At each level j , signed contributions to a single entry `vpathsumsj+1[ $u_2$ ][ $v_2$ ]` arise only from upstream pairs $(u', vp), (u'', vp')$ that lie in the same Bruhat fiber of `monopermj` above v_2 . In particular, the cancellation blow-up factor at level j is bounded by the maximum size of a length- θ_j `kdown`-fiber, i.e. by*

$$F_j := \max_{v_2} \#\{vp : vp \xrightarrow{\leq \theta_j \text{ Bruhat descents of } \text{monoperm}_j} v_2\} \leq \binom{\ell(\text{monoperm}_j)}{\theta_j}.$$

Proof. Sign-flips in `kdown_perms` are produced exclusively by Bruhat descents in the auxiliary chain from `monopermj`. Two upstream contributions to a given (u_2, v_2) that carry opposite signs must therefore differ by an even/odd parity in the number of `monopermj`-descents taken. In particular both pre-images vp, vp' of v_2 lie inside a single Bruhat fiber of length $\leq \theta_j$, of size at most $\binom{\ell(\text{monoperm}_j)}{\theta_j}$. $\square$

Theorem 5.1. *The running time of `schubmult.py` on (u, v) satisfies*

$$T_{\text{schub}}(u, v) \leq \text{poly}(n) \cdot M_{\text{out}}(u, v) \cdot \prod_{j=1}^r F_j,$$

where $M_{\text{out}} = \sum_w c_{u,v}^w$ is the total multiplicity and F_j is the per-level cancellation fiber bound of Proposition 5.3.1. In particular, if all parts θ_j are bounded by a constant, the cancellation overhead is at most a polynomial in n per level, giving

$$T_{\text{schub}}(u, v) = \tilde{O}\left(M_{\text{out}}(u, v) \cdot n^{O(r)}\right).$$

5.4. What does “dense multiplication” mean? Specifying the problem. Comparing `schubmult.py` against “dense” multiplication only makes sense once both the input and output bases are pinned down, since basis changes between the monomial basis and the Schubert basis are themselves nontrivial. We distinguish four problems:

- (P1) Mon→Mon: given $\mathfrak{S}_u, \mathfrak{S}_v$ as polynomials in the Artin monomial basis, output their product in the monomial basis.
- (P2) Sch→Mon: given $u, v \in S_n$, output $\mathfrak{S}_u \cdot \mathfrak{S}_v$ in the monomial basis.
- (P3) Mon→Sch: given $\mathfrak{S}_u, \mathfrak{S}_v$ as polynomials, output the structure constants $c_{u,v}^w$.
- (P4) Sch→Sch: given $u, v \in S_n$, output the structure constants $c_{u,v}^w$. *This is the problem `schubmult.py` solves.*

The four problems differ by basis-change steps. Calling any two of them “the same problem” is sloppy: each basis change is itself a nontrivial algorithm, and the dominant case dramatically illustrates the difference. If v is dominant, $\mathfrak{S}_v = x^{\text{code}(v)}$ is a single monomial, so problem (P1) is trivial — just shift the exponents of $\mathfrak{S}_u$ — while problem (P4) must additionally re-expand the result in the Schubert basis, which is genuine work and what `schubmult.py` does.

The fair baseline for (P4). Any honest competitor to `schubmult.py` must also output structure constants. The natural “dense” competitor for (P4) is

(MS) “*Multiply-then-straighten*”: solve (P2) by monomial multiplication, then convert the result to the Schubert basis via repeated divided differences (or any straightening algorithm).

Let $S_w := |\text{supp}(\mathfrak{S}_w)|$. The two stages of (MS) cost:

- Stage 1 (multiply): $S_u \cdot S_v$ monomial products plus dictionary operations, producing a polynomial of support size at most $A(n, \ell(u) + \ell(v))$.
- Stage 2 (straighten): expand the resulting polynomial of degree $\ell(u) + \ell(v)$ in the Schubert basis. The standard divided-difference algorithm $c_{u,v}^w = \partial_{w^{-1}w_0}(\mathfrak{S}_u \mathfrak{S}_v \cdot \mathfrak{S}_{w_0})|_{x=0}$ requires up to $\binom{n}{2}$ divided differences, each of which can roughly double the polynomial support, giving a worst-case stage-2 cost

$$T_{\text{straighten}}(u, v) \geq A(n, \ell(u) + \ell(v)) \cdot \text{poly}(n).$$

So in total

$$(5.4.1) \quad T_{\text{MS}}(u, v) \geq \max(S_u \cdot S_v, A(n, \ell(u) + \ell(v))) \cdot \text{poly}(n).$$

Both terms are necessary lower bounds: the first because the multiplication stage cannot be avoided, the second because any straightening algorithm must read every monomial of the intermediate polynomial.

Honest accounting in the dominant case. If v is dominant, $S_v = 1$ and stage 1 is trivial. But stage 2 of (MS) must straighten a polynomial of degree $\ell(u) + \ell(v)$ supported on a shifted Artin set of size $S_u \leq A(n, \ell(u))$. So $T_{\text{MS}} \geq A(n, \ell(u)) \cdot \text{poly}(n)$, while

$$T_{\text{schub}} = \tilde{O}(M_{\text{out}}(u, v))$$

by Corollary 5.2.2. The transition algorithm wins iff $M_{\text{out}}(u, v) \ll A(n, \ell(u))$. Since $M_{\text{out}}(u, v)$ is bounded by the number of pipe dreams for u shifted by $\text{code}(v)$ that survive the straightening relations, this ratio depends on u and is not uniformly favorable. *The dominant case is genuinely close, and we make no uniform claim there.*

5.5. Comparison and clean win regime. For arbitrary v , combining (5.1.1) with (5.4.1):

Theorem 5.2. *The transition algorithm beats the multiply-then-straighten baseline whenever*

$$(5.5.1) \quad \prod_{j=1}^r \binom{n+1}{\theta_j(v^{-1})} < \frac{A(n, \ell(u) + \ell(v))}{\text{poly}(n)},$$

i.e. whenever the rigorous `vpaths` fan-out is dominated by the size of the intermediate dense polynomial.

The Mahonian sequence $A(n, \cdot)$ is strongly log-concave, with peak $A(n, \binom{n}{2}/2) = \Theta(n!/n^{3/2})$ and Gaussian tail

$$A(n, d) = \frac{n!}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(d - N/2)^2}{2\sigma^2}\right) (1 + o(1)), \quad N := \binom{n}{2}, \quad \sigma^2 = \frac{2n^3 + 3n^2 - 5n}{72}.$$

Proposition 5.5.1. *Suppose $\theta(v^{-1})$ has all parts bounded by a constant c , and $\ell(u) + \ell(v)$ lies within $O(\sigma) = O(n^{3/2})$ of the Mahonian peak. Then*

$$\frac{T_{\text{MS}}(u, v)}{T_{\text{schub}}(u, v)} \geq \frac{n!}{n^{O(cr)}} = \frac{n!}{\text{poly}(n)},$$

i.e. `schubmult.py` beats multiply-then-straighten by a factor $\sim n!$ in this regime.

Proof. Under the part-size hypothesis, $\prod_j \binom{n+1}{\theta_j} \leq (n+1)^{cr}$. The Mahonian estimate gives $A(n, \ell(u) + \ell(v)) = \Theta(n!/\sigma) = \Theta(n!/n^{3/2})$ in the stated window. Substituting into (5.5.1) yields the claim. $\square$

Remark. The win factor of Proposition 5.5.1 comes entirely from *not* forming the intermediate polynomial $\mathfrak{S}_u \cdot \mathfrak{S}_v$ in the monomial basis. The transition algorithm avoids ever writing down a polynomial of size $A(n, \ell(u) + \ell(v))$ — its intermediate state is the Schubert-coefficient dictionary, which under the bounded-parts hypothesis has size only $\text{poly}(n)^r$. The basis change is performed *incrementally and selectively*, level by level through `kdown_perms`, rather than as a single monolithic straightening at the end.

The picture that emerges is that the transition algorithm avoids both the monomial blow-up of the intermediate product and the cost of a full straightening pass; cancellation overhead, when present, is fiber-confined. The clean unconditional win regime is bounded part-sizes in $\theta(v^{-1})$ with $\ell(u) + \ell(v)$ near the Mahonian peak; the dominant case is *not* a clean win and we explicitly do not claim it.

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